# Harpuneet Singh

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## Education

**University of Toronto** Aug 2023 – Dec 2028
*HBSc Computer Science (Co-op) · Toronto, Ontario, Canada*

**Learning Paths High School** Jul 2021 – May 2023
*High School Diploma · Ajitgarh, Punjab, India*

**St Kabir Public School** Jul 2011 – Jul 2021
*Secondary School Diploma · Chandigarh, India*

## Honours and Awards

Laidlaw Research Scholar (one of 25 scholars; $12,500 funding); PRISM Fellow; Dean's List Scholar; Dr. Babak Farzad's Award of Leadership. First place, UN Transformative Research Challenge ($5,000 funding).

## Research Experience

**SpectraSip** May 2026 – Present
*Interactive Intelligent Systems Laboratory · Tokyo, Japan*

- Developing a low-cost smart coaster that identifies near-infrared spectral fingerprints of adulterated drinks.
- Ran 180+ controlled trials across drug concentrations, drink compositions, and lighting conditions. Trained a multiband NIR classifier with 92% adulterant detection accuracy.
- Deployed the system in Tokyo nightlife venues and reduced ambient-light interference by 75% across four hardware iterations.

**Opto-Dye** Mar 2025 – Apr 2026
*Wheeler Microfluidics Laboratory · Toronto, Canada*

- Designed a microfluidic–optical system for low-resource forensic hyperspectral microscopy across 350–1080 nm, working with Prof. Aaron Wheeler and the Ontario Centre for Forensic Sciences.
- Developed a deep-learning classifier using Detectron2 and $\mu$SAM for unstained, multichannel microscopy samples. Integrated imaging with a microfluidic DNA-analysis chip, reducing sample processing time by 70%.
- Validated on 200+ samples with 97% accuracy against traditional staining and hyperspectral methods. Created a portable, low-power imaging unit 50 times cheaper than comparable commercial systems.
- Supported by $12,500 in Laidlaw research funding; selected as one of 25 scholars.

**ChromaFlash** Apr 2024 – Dec 2025
*Computational Health & Interaction Laboratory · Toronto, Canada*

- Built a programmable LED flash that adapts illumination to improve optical contrast between materials, under Prof. Alex Mariakakis. The work explored spectral imaging with ubiquitous cameras.
- Characterized 50+ materials across 400–900 nm. Designed high-speed PCBs in Altium with 7 ns LED switching and implemented per-channel brightness and timing control on a Raspberry Pi Pico 2.
- Created a gradient-based optimization algorithm for spectral separability. Achieved 13 dB material contrast, 10 nm illumination granularity, and 800 lumens of scene brightness.

## Research Experience (continued)

**Cognitive Restructuring** Aug 2024 – Aug 2025
*Intelligent Adaptive Interventions Laboratory · Toronto, Canada*

- Selected as one of four undergraduates for research with Prof. Joseph J. Williams. Developed an LLM-based cognitive restructuring framework using Python, LangChain, Keras, and TensorFlow.
- Optimized the model architecture for a 40% improvement in inference efficiency and a 30% reduction in processing latency.
- Redesigned the interface using React and Node.js, increasing the daily user base by 200 users. Also used Django.

**AgroArmor** Apr 2024 – Nov 2024
*United Nations Innovation Laboratory · Rome, Italy*

- Designed a scalable, modular biogas plant for rural agricultural settings.
- Engineered custom PCBs and IoT sensor modules for low-power slurry agitation, automated gas-flow control, and real-time monitoring. Prototyped in Fusion 360 and AutoCAD, with MATLAB models of gas flow and power consumption.
- Integrated eco-friendly materials to reduce costs by 30% while supporting durability and repairability. Won first place in the UN Transformative Research Challenge and secured $5,000 in research funding.

**Mersivity / SWIM** Oct 2023 – Jan 2024
*Mannlabs · Toronto, Canada*

- Worked with Prof. Steve Mann on inclusive VR technology for people with spinal muscular atrophy, using an EEG-based direct brain interface.
- Built three Sequential Wave Imaging devices that create a VR environment from a single LED strip. Prototyped a custom EasyEDA PCB with addressable LEDs controlled by an ESP32 Feather.
- Integrated a Muse InteraXon EEG headband, using Python and C++ to translate sensor readings into VR actions and visible wave forms.

## Entrepreneurship and Industry Experience

**STEMabled** May 2022 – Aug 2023
*Founder · Accessible education initiative*

- Founded an initiative developing inclusive Arduino STEM kits for students with disabilities. Designed and 3D-printed four kits with Braille markings using TinkerCAD.
- Distributed and tested kits in collaboration with IIT Delhi, Plaksha University, and the Institute for the Blind.
- Led C++ and MicroPython workshops for 150+ visually impaired students and trained 16 faculty members. Developed audio and Braille training modules to support independent exploration.

**Electric Vehicle Forecasting** May 2023 – Jul 2023
*Tendril Solutions Research · Research Intern*

- Built predictive models for electric-vehicle valuations using Python, Keras, and TensorFlow. Compared social platforms to assess the usefulness of brand sentiment.
- Improved data collection by parsing Reddit and Twitter discussions. Reported 94% model accuracy and an $R^2$ score of 0.92, with 18% less processing time and 10% lower token cost.

## Earlier Projects and Research

**EzOCR** Dec 2021 – Mar 2022
*Learning Paths and IBM*

- Partnered with IBM to develop an OCR app to help teachers detect plagiarism.
- Adopted by 40+ faculty, with an estimated 80% reduction in academic dishonesty.

**SuprStudent Security Testing** Oct 2021 – Nov 2021
*SuprStudent · Cybersecurity Intern*

- Performed ethical security testing on SuprStudent's education applications and websites.
- Identified, documented, and resolved 17 bugs, glitches, and security risks.

**Young Research Fellow** Jun 2022 – Jul 2022
*Shoolini University*

- Selected as one of 20 students nationwide; shadowed professors in statistics, data science, physics, and food science.
- Contributed to filing three patents for researchers, with 200+ hours of research in patent law.

## Leadership, Service and Student Projects

**UofTHacks – Sponsorship Executive** May 2025 – Present
*University of Toronto*

Led sponsor outreach, contacted 100+ sponsors, pitched initiatives, and secured funding for the university's flagship hackathon.

**USRP – Research Mentor** Jan 2025 – Present
*Undergraduate Student Research Program*

Organizing workshops and mentoring 10 undergraduates in project development, skills, and research opportunities.

**UofT Blueprint – Software Developer** Sep 2024 – May 2025
*Period Purse*

Developed an Android app for a menstrual equity charity using Kotlin, Coroutines, and Android Jetpack.

**UTMIST – Project Developer** Aug 2024 – Aug 2025
*Lovelytics task automation*

Developed an AI-based automation pipeline using Python, Hugging Face, TensorFlow, JavaScript, and Node.js.

## Publications and Posters

**Singh H**, Verma D, Mariakakis A. *Multispectral LED microscopy for rapid, dye-free sperm cell identification in forensic samples.* F1000Research 2026;15:311. Conference poster, presented at the Laidlaw Scholars Annual Conferences 2025. doi:10.7490/f1000research.1120482.1.

## Technical Skills

**Languages:** C++, Python, C, Verilog, MicroPython, Assembly, R, MATLAB, Java, MySQL, HTML, JavaScript, Kotlin, Git
**Prototyping:** Arduino, Raspberry Pi, IoT Sensors, AdaFruit, SolidWorks, AutoCAD, Fusion, EasyEDA, Altium, Logisim
**Equipment:** Fluorescence microscope, Optical spectrometer, Microscope cameras, Tunable filters, Dichroics, Centrifuge, Gel electrophoresis, Thermocycler, Piezo stages, Oscilloscope, Network analyzer, FPGAs, Reflow oven, 3D printers
**Network Interfaces:** LoRaWAN, Wi-Fi, Bluetooth LE, NFC/RFID, USB, Ethernet, MQTT, $I^2C$, SPI, UART, CAN
**Frameworks and libraries:** Angular, Django, Espresso, Flutter, Node.js, React, Flask, LangChain, TensorFlow, Keras, PyTorch, Scikit-learn, Matplotlib, Android Jetpack, Apache, Coroutines, Guava, Pandas, PySpark, Hugging Face, Detectron2, $\mu$SAM.